Linear Algebra (MT-1004)

Homework#11

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Question 1

Consider the system of differential equations

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}, \quad \text{where } A = \begin{bmatrix} 4 & 1 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad \mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}.$$

Solve the system using the idea of eigenvalues and eigenvectors.